

JOHN MARTIN MBURU NKONGE

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I am an IT Specialist passionate about Cloud Computing and Software Development with hands-on experience in designing, building, and maintaining cloud-native applications that solve real business problems. I am focused on building robust and scalable Fintech, Edtech and Sustainable solutions, API development, and Cloud Infrastructure.

WORK EXPERIENCE

Fullstack Engineer/Tech Lead — Dubu Huru Ltd (Remote) August 2025 – Present

- Led a cross-functional team of 4 (frontend, cloud, and cybersecurity engineers) in the architectural design, development, and deployment of a highly available Fintech SaaS (crowdfunding/fundraising) platform serving users across Africa and globally.
- Architected the backend using Python (Django, Django REST Framework) with a RESTful API design, achieving sub-millisecond to low-millisecond API response times and ~90% uptime in early production.
- Implemented redundant and fault-tolerant infrastructure using AWS ECS Fargate with Auto Scaling and Application Load Balancer achieving 99.9% system availability across multiple Availability Zones in `af-south-1` (Cape Town) with an active failover region in `us-east-1` (Virginia).
- Optimized database performance using AWS RDS PostgreSQL with Read Replicas and implemented ElastiCache (Redis) for session storage and Celery task brokering.
- Decoupled slow operations like payment callbacks, webhook processing, and heavy analytics using Celery workers with Redis as the message broker, successfully processing 200+ transactions in early production.
- Integrated multi-currency payment solutions (M-Pesa, Stripe, PayPal, Payoutstack) with strict input validation, idempotency handling, and webhook signature verification.
- Optimized asset delivery using AWS CloudFront as a CDN for media files served from S3, reducing latency for African users.
- Configured secure multi-tier networking using VPC with private subnets, NAT Gateway, and security groups to isolate backend services and RDS from public internet exposure.
- Built and maintained CI/CD pipelines for both staging and production environments using GitHub Actions, with multi-stage Docker builds to optimize image size and push to AWS ECR for ECS Fargate deployments.
- Deployed a React frontend on AWS Amplify (`us-east-1`) with custom domain routing via Route 53 failover policy and SSL termination at the ALB using ACM certificates.

- Monitored and observed production systems using Prometheus and AWS CloudWatch, with error tracking and alerting via Sentry.
- Managed infrastructure as code using Terraform, enabling reproducible and version-controlled environment provisioning across staging and production.

Software Engineer — JamboPay (Nairobi) May 2024 – August 2025

- Developed and maintained a multi-tenant County E-Revenue Management System serving 10 counties across Kenya (Embu, Tharaka Nithi, Busia, Kisii, Uasin Gishu, Bomet, Kajiado, Samburu, Meru, and Samburu National Park), handling revenue streams across parking fees, land rates, business permits, and e-wallet payments.
- Led the migration of legacy Django MVT modules to a decoupled Django REST Framework (DRF) architecture with Vanilla JS frontends, reducing frontend-to-backend latency by 30% and enabling independent scaling and maintenance of each revenue module.
- Designed and implemented new feature modules end-to-end — from PostgreSQL schema design and API development to Vanilla JS dashboard integration — across e-wallet, USSD payment, and county revenue collection flows.
- Improved revenue reporting accuracy by 15% by implementing precise geospatial mapping using PostGIS and AnyChart JS, enabling location-based revenue tracking and visualization for county administrators.
- Improved database query efficiency by 22% through schema normalization, composite indexing on high-traffic multi-tenant revenue tables, and query profiling — reducing filtered data retrieval for county administrators from seconds to milliseconds across large cross-county datasets.
- Decreased production bug reports by 20% by introducing a standardized unit testing and E2E testing suite covering critical e-wallet and USSD payment flows, establishing testing as a core part of the development workflow.
- Diagnosed and resolved defects in reporting modules for a government revenue system serving active county operations, ensuring data integrity and compliance with county financial reporting requirements.

Software Developer Intern — JamboPay (Nairobi) May 2023 – August 2023

- Learned Python programming and Django/Django REST Framework.
- Developed backend project assignments assigned by the CTO.
- Collaborated with developers to ensure smooth project development and requirements achievement.

EDUCATION

Bachelor of Science in Information Technology — Mount Kenya University *Graduated: 2023*

High School Diploma (KCSE) — Lenana School *Graduated: 2019*

LANGUAGES

- **English:** Fluent
 - **Kiswahili:** Native
 - **German:** A1 Level
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TOOLS & TECH STACK

- **Backend:** Python, Django, Flask, Fast API
 - **Frontend:** React TS/JS, JavaScript, HTML5, CSS3, Angular JS
 - **Cloud/DevOps:** AWS (S3, EC2, Lambda, VPC, IAM, RDS, Amplify), Docker, Kubernetes, CI/CD, Cloudwatch
 - **Databases:** PostgreSQL, PostGIS, MySQL, SQLite, Redis
 - **Other:** Kafka, RabbitMQ, Git, AI/LLM APIs
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CERTIFICATIONS

- **KCNA:** Kubernetes and Cloud Native Associate
 - **AWS:** Cloud Practitioner, DevOps on AWS, Cloud 101
 - **Networking:** CISCO CCNA
 - **Data:** Big Data
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SOFT SKILLS

- Problem-solving and clean code delivery
- Team leadership and mentoring
- Cross-team collaboration and technical communication